

CSCI 455 – Robotics

Final Project: Autonomous Robot Greeter for the new building

Due Dates

- **Final Demo:** (Anytime before the end of class on April 29th)
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Overview

In this final project, your robot will act as an **autonomous greeter assistant**.

Starting from a fixed location, the robot will:

1. Detect a human
2. Greet them
3. Accept a request (bathroom or robot lab) using speech recognition (no typing)
4. Navigate autonomously using LIDAR
5. Guide the user to the correct destination

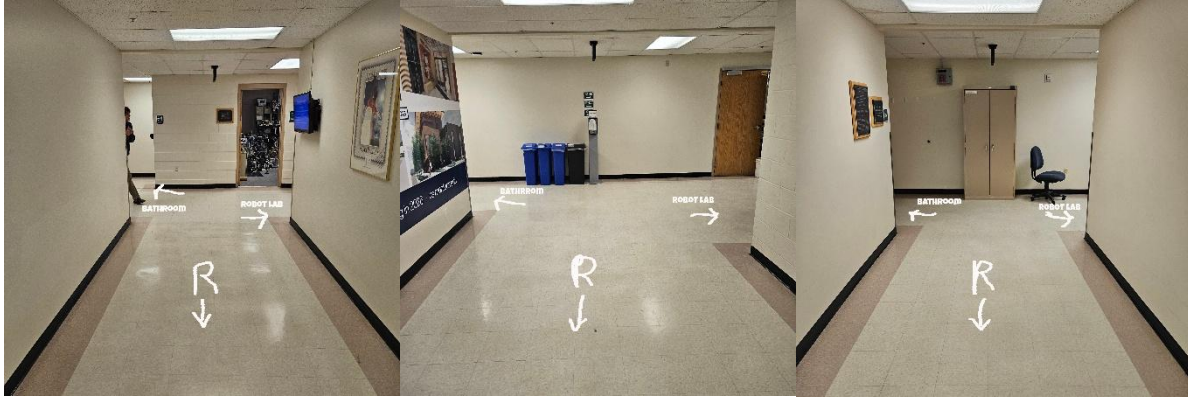
This project integrates:

- LIDAR sensing
- Autonomous navigation (wall following)
- Obstacle avoidance
- Speech input
- Finite State Machine design

System Behavior

In the hallway example images, the robot starts where the **R** is located, facing in the direction of the arrow. A human will approach from the front. When the robot detects the human, it should greet them and ask how it can help. The human will say either, “I need to go to the bathroom” or “I need to go to the robot lab.” The robot will then say, “Follow me.”

The robot must then turn around, follow the hallway wall to the intersection, turn in the correct direction, continue for about 5 more feet, and say, “We have arrived at the bathroom” or “We have arrived at the robot lab.”



Starting Condition

- Robot begins at the **START position**, facing forward (as shown in map images of our hallway outside.)
- Robot is idle, waiting for a human

Interaction Sequence

1. Human Detection

Robot waits until a human is detected using the lidar detection of a human approaching.

2. Greeting

Robot says:

“Hello, how can I help you?”

3. Command Input

User must provide one of:

- “I am looking for the bathroom”
- “I am looking for the robot lab”

Speech Requirement : The robot must take in speech recognition and parse the speech to the proper command of lab or bathroom

You must implement ONE of:

Option A (Full credit)

- Speech recognition (online or local)

Option B (Minus 15 points)

- Keyboard input (typed command)

You must clearly state which option you use.

The Bluetooth wireless speakers have built-in microphones

4. Robot Response

Robot says:

“Follow me.”

Navigation Requirements

Phase 1: Initialization

Robot must:

- Turn **180 degrees**
- Move forward until it detects **walls on both sides** (hallway alignment)

Phase 2: Move to T-Intersection

Robot moves forward using:

- Wall following
- Obstacle avoidance (If I step in front the robot should stop).

T-Intersection Detection

The robot detects the T by:

- A wall in front within threshold distance , and wall missing on the left and right.
 - The wall in front should be different than an obstacle in the way, you can rely on no bad obstacles blocking the way after you get to the intersection.
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Phase 3: Decision Turn

Based on command:

Command Action

Bathroom Turn RIGHT

Robot Lab Turn LEFT or continue (based on your map implementation)

Phase 4: Final Navigation

After the turn:

- Continue straight for **5 seconds**
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Phase 5: Stop

Robot stops and will say:

“We have arrived at the xxxxx.” (lab or bathroom)

Obstacle Handling

At ALL TIMES:

- If obstacle detected in front:
 - Robot must stop OR adjust path
- Robot must NOT collide with walls or objects

Required Architecture

Finite State Machine (REQUIRED)

Your program must use a state-based system.

Minimum states:

WAITING

GREETING

LISTENING

TURNING_AROUND

ALIGNING_TO_HALLWAY

MOVING_TO_T

TURNING_TO_DESTINATION

FINAL_MOVEMENT

STOPPED

You must:

- Clearly define states
- Show transitions in code or comments

LIDAR Requirements

You must use LIDAR to:

- Detect front obstacles
- Detect side walls
- Detect intersections/openings

Wall Following Requirement

Robot must:

- Maintain approximate distance from wall, trying to stay centered in hallway.
- Adjust steering based on side distance

Implementation Notes (IMPORTANT)

Speech + Network Constraint

Because of Speech Recognition network requirements:

- You MAY:
 - Need to have your robot connected to a live phone hotspot
 - OR
 - Connected to MSU-Guest

OR

- Use keyboard input if you want to forfeit the 15 points

Testing Environment

Because multiple robots cannot use the same hallway try not to bother the classrooms down the hallway, go the other direction. And I have some robot cases in the lab that can be arranged as walls.

What to Turn In

1. Live Demo (Required)

Robot must:

- Complete full interaction + navigation

2. Project Log (Required)

Include:

- Description of your FSM
- Explanation of wall-following logic
- How you detect intersections
- Issues encountered and fixes

Grading Rubric (100 points)

Category	Points
Human interaction (greeting + input)	15
Speech Recognition	15
FSM implementation	10
Wall following behavior	20
Intersection detection and turning	15
Obstacle avoidance	10
Successful navigation to correct destination	15
	100
PLUS	
project log	25